

Element A:

Presentation and Justification of the Problem

Problem Statement: An engineering teacher decides on Thursday night to take his family on a spontaneous vacation over the weekend, leaving on a flight on Friday evening and getting back home on Sunday evening on a different flight. The teacher finds a desirable location to take the family and finds several activities for the family to enjoy including: a sporting event, a children's museum, an art gallery, and a laser tag arena. Upon breaking this wonderful news to the family, a problem is quickly discovered. One of the three children is young and has a difficult time locking public restroom doors. To make the situation worse, it is also known that the child has trouble using the restroom when other people are watching.

Since it is impossible to ensure that every stall in every bathroom has an adequate door, the family requires a solution that takes an ordinary bathroom stall without a lock and to secure the door closed.

According to Cleveland Clinic, Psychologically, trouble using the bathroom in front of others stems from social anxiety, involving a fear of judgment, embarrassment, and scrutiny over sounds, smells, or perceived incompetence. Unfortunately, there will not be enough time before the trip's departure to work through these mental issues.

The child will have to face this issue everywhere they go, and multiple times per day (specifically the number of times they must use the restroom). This problem is quite troublesome, since without a solution, the child will not be able to use the restroom, and will risk bladder infection or rupture. So, the family will not be able to go on the trip.



A solution to the lack of locks on many bathroom doors is a specially designed Binder Clip. This clip will be versatile enough to close almost all stall doors this family encounters. The effective binder clip device will be lightweight, small, durable, and reusable. To achieve this, the design will use a hard and sturdy plastic material with metal wires.

The primary stakeholders include the Child using the restroom and the parents of the child. The Binder Clip must be able to make the child feel comfortable enough to use the restroom and durable, compact, and lightweight enough as to not inconvenience the parents.

The secondary stakeholders are other patrons of the bathroom. Their wait times for using the restroom will be affected by how long the lock takes to use. Additionally, they depend on the lock working in order to not see the child using the restroom inside the stall.

The tertiary stakeholder is the hotels, restaurants, and airlines that the family would use on their vacation. These establishments would lose money that they would have made from the family trip if the lock is not effective enough for use.

In conclusion, without a reliable and reusable device to use on universal bathroom doors, the child will not be able to use the bathroom leading the family to not attend their weekend getaway.

<https://my.clevelandclinic.org/health/diseases/22208-shy-bladder-syndrome-paruresis>

<https://www.piedmont.org/blog/how-long-is-it-safe-to-hold-your-urine>